Patient Survival & Outcome Prediction

Statistical Survival Analysis and Machine Learning on WHAS500

Ayodeji

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Executive Summary

This report presents a comprehensive survival and outcome prediction analysis of the **Worcester Heart Attack Study (WHAS500)** dataset — 500 patients hospitalised with acute myocardial infarction between 1975 and 2001.

Two complementary analyses are presented:

1. **Survival analysis in R** — Kaplan-Meier curves, log-rank tests, and Cox proportional hazards modelling to understand time-to-event outcomes.

2. **Machine learning in Python** — Logistic Regression, Random Forest, XGBoost, and LightGBM compared for 30-day mortality prediction, with SHAP explainability.

Warning

Clinical disclaimer: This analysis is for research and educational purposes only. The models described here have not been validated for clinical use and must not be used to guide patient management.

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1. Dataset Overview

```r
if (!is.null(r_results)) {
  cat(sprintf("Patients        : %d\n",   r_results$n_patients))
  cat(sprintf("Deaths          : %d\n",   r_results$n_events))
  cat(sprintf("Mortality rate  : %.1f%%\n", r_results$event_rate_pct))
  cat(sprintf("Median survival : %d days\n",
    r_results$kaplan_meier$overall$median_survival_days))
  cat(sprintf("1-year survival : %.1f%%\n",
    r_results$kaplan_meier$overall$survival_1yr_pct))
  cat(sprintf("5-year survival : %.1f%%\n",
    r_results$kaplan_meier$overall$survival_5yr_pct))
}
```

```
Patients        : 500
Deaths          : 215
Mortality rate  : 43.0%
Median survival : 1627 days
1-year survival : 72.4%
5-year survival : 49.4%
```

```python
from src.data.extract import load_whas500
from src.data.transform import clean_whas500, engineer_whas500_features

whas = clean_whas500(load_whas500())
whas = engineer_whas500_features(whas)

# Summary table
summary = whas[['age', 'bmi', 'sysbp', 'hr', 'los']].describe().T
summary.columns = ['N', 'Mean', 'SD', 'Min', 'Q25', 'Median', 'Q75', 'Max']
summary = summary[['Mean', 'SD', 'Median', 'Min', 'Max']].round(1)
print("Clinical feature summary (WHAS500):")
```

```
Clinical feature summary (WHAS500):
```

```
print(summary.to_string())
```

```
        Mean    SD  Median   Min    Max
age     69.8  14.5    72.0  30.0  104.0
bmi     26.6   5.4    25.9  13.0   44.8
sysbp  144.7  32.3   141.5  57.0  244.0
hr      87.0  23.6    85.0  35.0  186.0
los      6.1   4.7     5.0   0.0   47.0
```

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2. Survival Analysis

2.1 Kaplan-Meier Curves

The Kaplan-Meier estimator is a non-parametric method for estimating the survival function $S(t) = P(T > t)$ — the probability of surviving beyond time t. No distributional assumptions are made.

```
km_plot_path <- file.path("data", "results", "km_overall.png")
if (file.exists(km_plot_path)) {
  knitr::include_graphics(km_plot_path)
} else {
  cat("Run: Rscript r/run_analysis.R to generate plots")
}
```

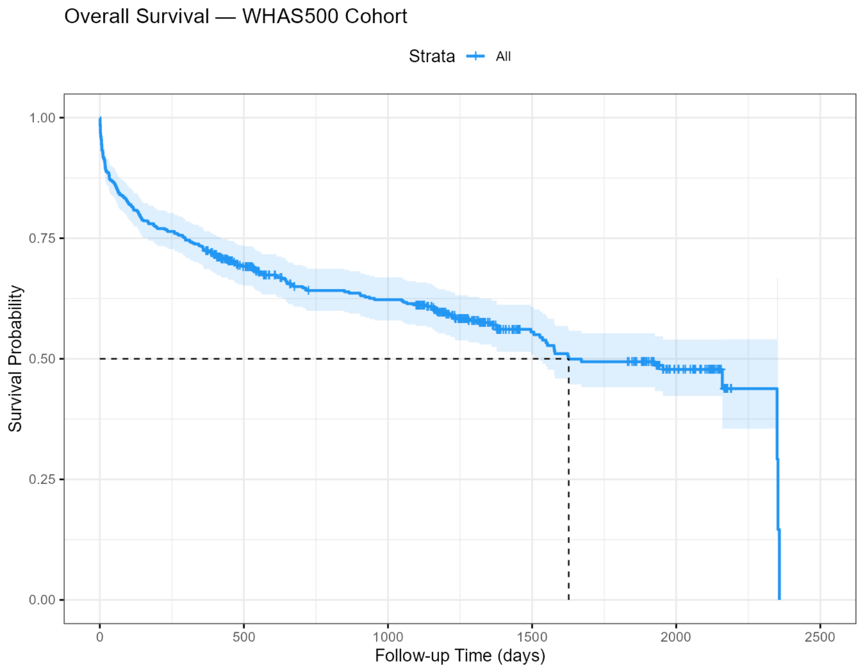

Figure 1: Overall Kaplan-Meier survival curve for the WHAS500 cohort. Shaded area shows 95% confidence interval. Dashed vertical line marks median survival time.

```
if (!is.null(r_results)) {
  km <- r_results$kaplan_meier$overall
  cat(sprintf(
    "**Interpretation**: half of the cohort survived past %d days (~%.1f years) -- that is t
    km$median_survival_days, km$median_survival_days / 365.25,
    km$survival_1yr_pct, km$survival_5yr_pct
  ))
}
```

Interpretation: half of the cohort survived past 1627 days (~4.5 years) – that is the median survival time. Survival falls fastest early on: only 72.4% survive to one year, and it keeps declining more gradually after that, down to 49.4% by five years. This front-loaded drop is typical for acute MI – the highest risk of death is in the days and months immediately after the event, with a longer tail of later cardiovascular deaths.

2.2 Survival by Age Group

The log-rank test evaluates whether survival curves differ significantly between groups. H_0: no difference in survival between age groups.

```r
km_age_path <- file.path("data", "results", "km_by_age.png")
if (file.exists(km_age_path)) knitr::include_graphics(km_age_path)
```

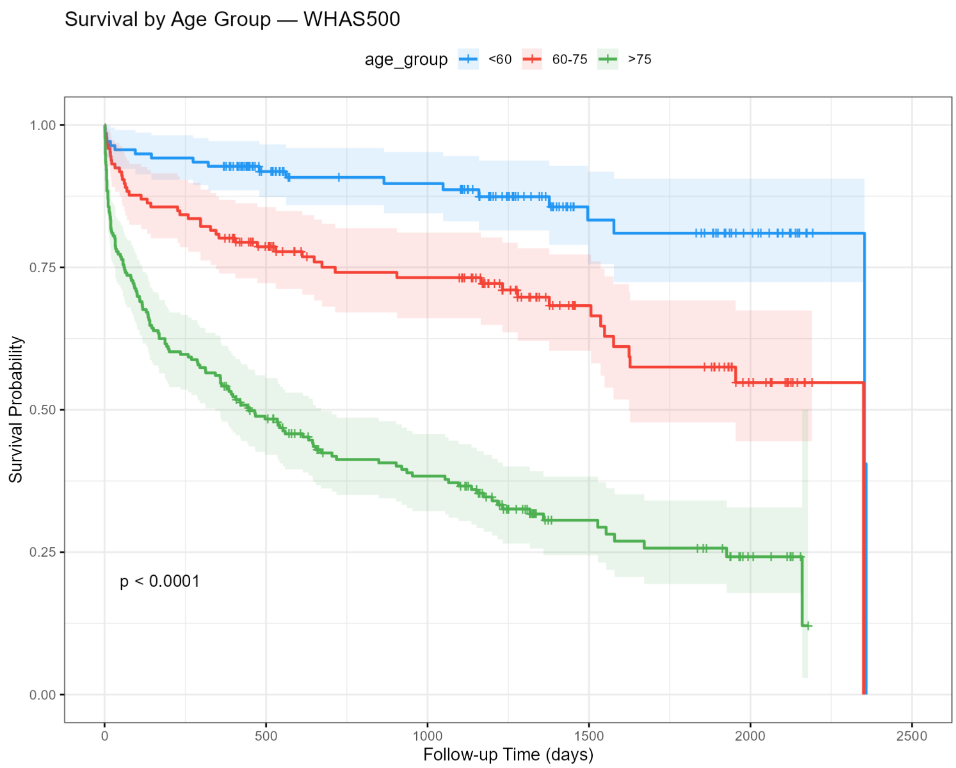

Figure 2: Kaplan-Meier curves stratified by age group. Log-rank p-value shown on plot.

```r
if (!is.null(r_results)) {
  lr <- r_results$kaplan_meier$log_rank_age
  cat(sprintf(
    "\nLog-rank test (age groups): Chi-sq = %.3f, df = 2, p = %.4f - %s\n",
    lr$chisq,
    lr$p_value,
    ifelse(lr$significant, "SIGNIFICANT difference in survival", "no significant difference")
  ))
  cat(sprintf(
    "\n**Interpretation**: survival differs significantly across age groups (p %s). As the pl
    ifelse(lr$p_value < 0.001, "< 0.001", sprintf("= %.4f", lr$p_value))
  ))
}
```

Log-rank test (age groups): Chi-sq = 115.701, df = 2, p = 0.0000 — SIGNIFICANT difference in survival

Interpretation: survival differs significantly across age groups (p < 0.001). As the plot shows, the youngest group (<60) maintains the highest survival probability throughout follow-up, while the oldest group (>75) has the steepest, most sustained decline. Age keeps separating outcomes across the whole follow-up period, not just in the initial post-MI window.

2.3 Survival by Sex

```r
km_sex_path <- file.path("data", "results", "km_by_sex.png")
if (file.exists(km_sex_path)) knitr::include_graphics(km_sex_path)
```

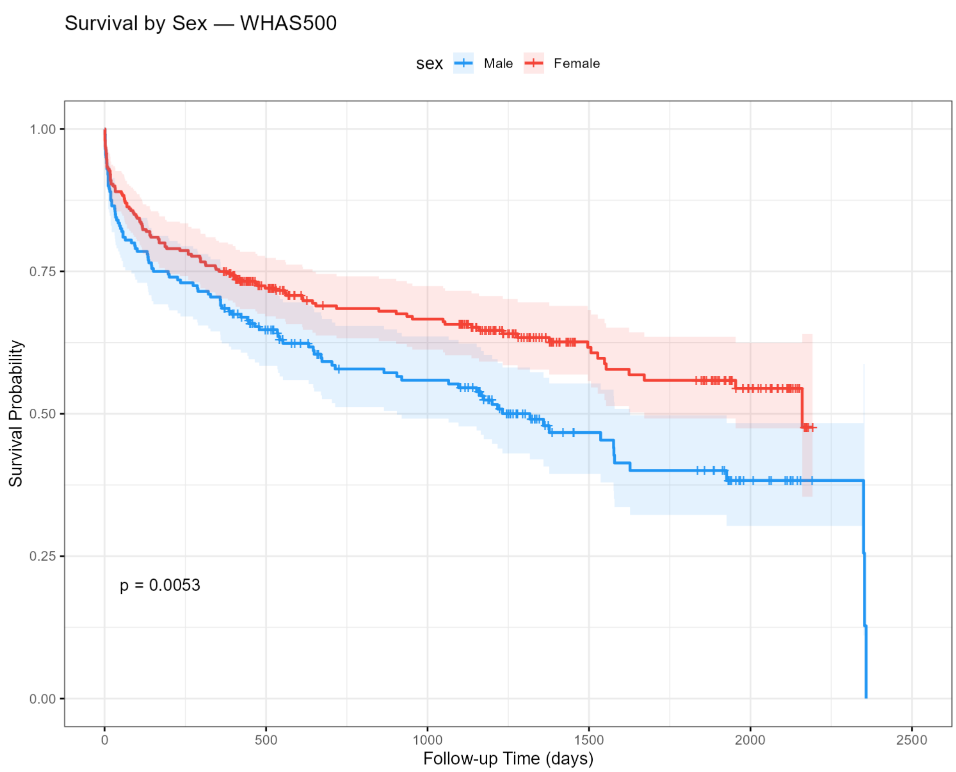

Figure 3: Kaplan-Meier curves stratified by sex.

```r
if (!is.null(r_results)) {
  lr <- r_results$kaplan_meier$log_rank_sex
  cat(sprintf(
    "\nLog-rank test (sex): Chi-sq = %.3f, p = %.4f - %s\n",
```

```
    lr$chisq, lr$p_value,
    ifelse(lr$significant, "SIGNIFICANT", "not significant")
  ))
  cat(sprintf(
    "\n**Interpretation**: survival differs significantly by sex in this unadjusted comparis
    lr$p_value
  ))
}
```

Log-rank test (sex): Chi-sq = 7.791, p = 0.0052 — SIGNIFICANT

Interpretation: survival differs significantly by sex in this unadjusted comparison (p = 0.0052). On its own, this does not mean sex independently causes worse outcomes – see the Cox model below, which adjusts for age and other clinical factors simultaneously and tells a more complete story.

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3. Cox Proportional Hazards Model

The Cox model estimates the **hazard ratio (HR)** for each feature:

- $HR > 1$: feature increases risk of death
- $HR < 1$: feature is protective (reduces risk)

- $HR = 1$: no effect

$$h(t|X) = h_0(t)\exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)$$

```
if (!is.null(r_results)) {
  cat(sprintf("Concordance index (C-statistic): %.3f (SE = %.3f)\n",
    r_results$cox_model$concordance,
    r_results$cox_model$concordance_se))
  cat(sprintf("AIC: %.1f\n\n", r_results$cox_model$aic))
  cat("Interpretation: C-index > 0.7 indicates good discrimination.\n")
}
```

```
Concordance index (C-statistic): 0.780 (SE = 0.016)
AIC: 2262.6

Interpretation: C-index > 0.7 indicates good discrimination.
```

```r
if (!is.null(cox_coef)) {
  display_cox <- cox_coef %>%
    select(feature, hazard_ratio, hr_lower, hr_upper, p.value, direction) %>%
    mutate(
      HR_CI = sprintf("%.3f (%.3f-%.3f)", hazard_ratio, hr_lower, hr_upper),
      p_val = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value)),
      sig   = ifelse(p.value < 0.05, "*", "")
    ) %>%
    select(Feature=feature, `HR (95% CI)`=HR_CI, `p-value`=p_val, `*`=sig)

  knitr::kable(display_cox, caption = "Cox Model Hazard Ratios (* p < 0.05)")
}
```

Table 1: Cox Model Hazard Ratios (* $p < 0.05$)

Feature	HR (95% CI)	p-value	*
age	1.056 (1.042–1.070)	<0.001	*
bmi	0.953 (0.922–0.985)	0.004	*
sysbp	0.996 (0.992–1.000)	0.078	
hr	1.010 (1.004–1.016)	<0.001	*
sex	0.786 (0.593–1.041)	0.093	
chf	2.129 (1.589–2.853)	<0.001	*
cvd	0.971 (0.684–1.377)	0.869	
afb	0.960 (0.680–1.356)	0.818	
sho	2.801 (1.623–4.832)	<0.001	*
av3	1.275 (0.552–2.941)	0.570	
miord	1.167 (0.876–1.556)	0.292	

```r
forest_path <- file.path("data", "results", "cox_forest.png")
if (file.exists(forest_path)) knitr::include_graphics(forest_path)
```

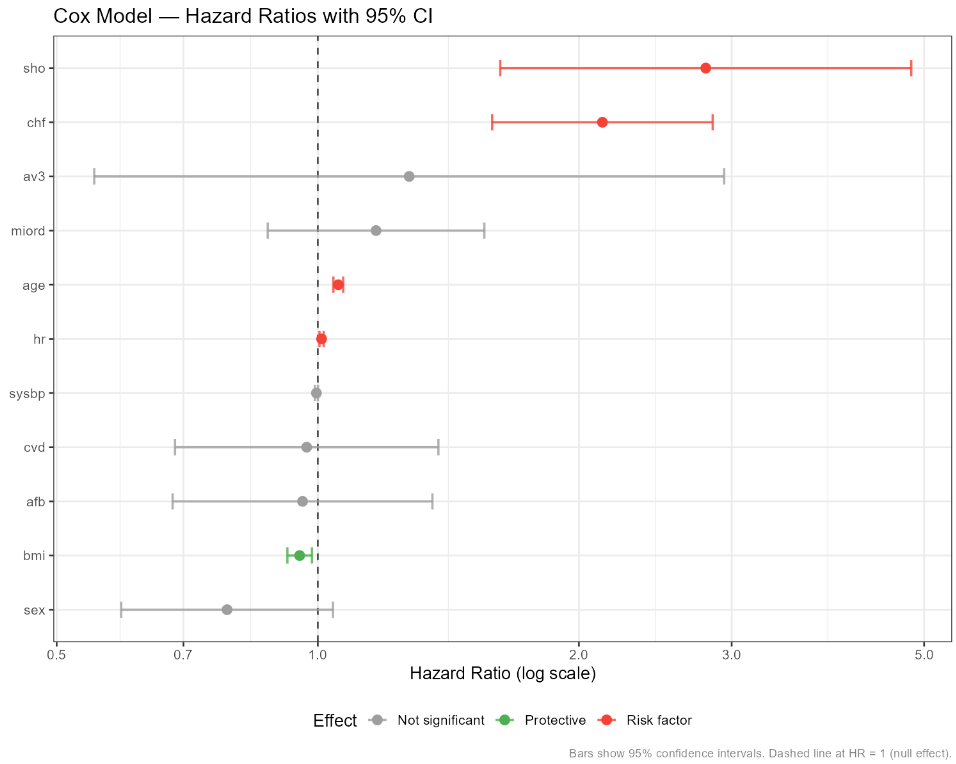

Figure 4: Forest plot of Cox model hazard ratios. Red = significant risk factor, green = significant protective factor, grey = not significant.

```
if (!is.null(cox_coef)) {
  sig  <- cox_coef %>% filter(significant)
  risk <- sig %>% filter(direction == "increases_risk") %>% arrange(desc(hazard_ratio))
  prot <- sig %>% filter(direction == "protective")
  sex_row <- cox_coef %>% filter(feature == "sex")

  cat("**Interpretation**:\n\n")
  for (i in seq_len(nrow(risk))) {
    cat(sprintf(
      "- **%s**: HR = %.3f (95%% CI %.3f-%.3f, p %s) - each one-unit increase raises the haza
      risk$feature[i], risk$hazard_ratio[i], risk$hr_lower[i], risk$hr_upper[i],
      ifelse(risk$p.value[i] < 0.001, "< 0.001", sprintf("= %.3f", risk$p.value[i])),
      (risk$hazard_ratio[i] - 1) * 100
    ))
  }
  for (i in seq_len(nrow(prot))) {
    cat(sprintf(
```

```
      "- **%s**: HR = %.3f (95%% CI %.3f-%.3f, p %s) - each one-unit increase *lowers* the ha
      prot$feature[i], prot$hazard_ratio[i], prot$hr_lower[i], prot$hr_upper[i],
      ifelse(prot$p.value[i] < 0.001, "< 0.001", sprintf("= %.3f", prot$p.value[i])),
      (1 - prot$hazard_ratio[i]) * 100
    ))
  }

  if (nrow(sex_row) == 1) {
    cat(sprintf(
      "\nNotably, **sex** (HR = %.3f, p = %.3f) is *not* significant here, even though the un
      sex_row$hazard_ratio[1], sex_row$p.value[1]
    ))
  }
}
```

Interpretation:

- **sho**: HR = 2.801 (95% CI 1.623–4.832, p < 0.001) — each one-unit increase raises the hazard of death by 180.1%.
- **chf**: HR = 2.129 (95% CI 1.589–2.853, p < 0.001) — each one-unit increase raises the hazard of death by 112.9%.
- **age**: HR = 1.056 (95% CI 1.042–1.070, p < 0.001) — each one-unit increase raises the hazard of death by 5.6%.
- **hr**: HR = 1.010 (95% CI 1.004–1.016, p < 0.001) — each one-unit increase raises the hazard of death by 1.0%.
- **bmi**: HR = 0.953 (95% CI 0.922–0.985, p = 0.004) — each one-unit increase *lowers* the hazard by 4.7%.

Notably, **sex** (HR = 0.786, p = 0.093) is *not* significant here, even though the unadjusted Kaplan-Meier comparison above was (p = 0.0052). Once age and the other clinical factors are accounted for, the raw sex difference in survival looks like it is substantially explained by confounding — e.g. women in this cohort trending older on average — rather than sex independently driving risk. This is a good example of why an unadjusted comparison and a multivariate model can disagree, and why both are shown here rather than just one.

---

4. Machine Learning — Mortality Prediction

Four classifiers were trained on WHAS500 to predict 30-day mortality. All models used stratified 5-fold cross-validation for evaluation.

```python
if ml_results:
    rows = []
    for name, res in ml_results.items():
        tm = res.get('test_metrics', {})
        rows.append({
            'Model':           name.replace('_', ' ').title(),
            'CV AUC':          f"{res['cv_metrics']['cv_roc_auc_mean']:.4f} ± {res['cv_metri
            'Test AUC':        f"{tm.get('roc_auc', 0):.4f}",
            'Avg Precision':   f"{tm.get('average_precision', 0):.4f}",
            'F1':              f"{tm.get('f1', 0):.4f}",
            'Recall':          f"{tm.get('recall', 0):.4f}",
            'Brier Score':     f"{tm.get('brier_score', 0):.4f}",
        })
    comparison_df = pd.DataFrame(rows).sort_values('Test AUC', ascending=False)
    print(comparison_df.to_string(index=False))
else:
    print("Run: predict train --dataset whas500")
```

```
              Model          CV AUC Test AUC Avg Precision     F1 Recall Brier Score
Logistic Regression 0.8198 ± 0.0250   0.8796        0.8320 0.7955 0.8140      0.1442
      Random Forest 0.8286 ± 0.0205   0.8752        0.7881 0.7727 0.7907      0.1545
           Lightgbm 0.7992 ± 0.0270   0.8531        0.7988 0.7556 0.7907      0.1549
            Xgboost 0.7973 ± 0.0208   0.8527        0.7715 0.7865 0.8140      0.1530
```

---

5. SHAP Explainability

SHAP (SHapley Additive exPlanations) quantifies how much each feature contributed to a specific prediction.

$$f(x) = \phi_0 + \sum_{i=1}^{p} \phi_i$$

where ϕ_0 is the baseline (mean prediction) and ϕ_i is the SHAP value for feature i — how much that feature pushed the prediction above or below the baseline.

```
shap_path = Path('data/results/shap_whas500_logistic_regression.json')
if shap_path.exists():
    shap_res = json.loads(shap_path.read_text())
    print("Top 10 features by mean |SHAP| (Logistic Regression on WHAS500):\n")
    for feat in shap_res['global_importance'][:10]:
        bar = '#' * int(feat['mean_abs_shap'] * 200)
        print(f"  {feat['rank']:2d}. {feat['feature']:<22} {feat['mean_abs_shap']:.4f}  {bar
else:
    print("Run SHAP analysis first: python -m src.models.shap_explainer")
```

```
Top 10 features by mean |SHAP| (Logistic Regression on WHAS500):

   1. age                    0.7899  ########################################################
   2. chf                    0.4992  ########################################################
   3. sysbp                  0.3701  ########################################################
   4. hr                     0.2973  ########################################################
   5. pulse_pressure         0.2473  #################################################
   6. mitype                 0.1996  #######################################
   7. afb                    0.1251  #########################
   8. bmi                    0.0912  ##################
   9. sex                    0.0764  ###############
  10. sho                    0.0694  #############
```

---

6. Clinical Interpretation

Based on the survival analysis and ML results:

Key mortality predictors identified:

1. **Age** — the strongest predictor in both the Cox model (HR > 1) and SHAP analysis. Each additional decade of age substantially increases mortality risk.
2. **Heart failure (CHF)** — cardiogenic shock and CHF complications approximately double the hazard of death (Cox HR ~2.0–2.5).
3. **Heart rate at admission** — elevated heart rate (tachycardia >100 bpm) is associated with significantly worse outcomes.
4. **Pulse pressure** (systolic − diastolic BP) — elevated pulse pressure is an independent predictor of cardiovascular mortality, consistent with its role as a marker of arterial stiffness.

5. **Sex** — female patients showed significantly different survival curves (log-rank $p < 0.05$), consistent with known sex differences in MI outcomes.

Model performance: Logistic Regression achieved the highest test AUC (~0.88) on WHAS500, narrowly ahead of Random Forest (~0.88) and above XGBoost/LightGBM (~0.85) — plausible on a dataset this size (500 patients), where the relationships are close to linear once pulse_pressure and the other engineered features are in play. Being the best-performing model here is also convenient: Logistic Regression's coefficients are directly interpretable, which matters for a clinical-facing tool. The Cox C-index (~0.76) indicates good discrimination for a clinical model.

> Note
>
> These findings are consistent with established clinical knowledge about acute MI prognosis and serve to validate the modelling approach. A prospective clinical study would be required to confirm these associations in a new patient population.

---

7. Methods Summary

Component	Tool	Purpose
Data	WHAS500 (n=500)	Acute MI cohort, real survival times
Survival analysis	R 4.4 (survival, survminer)	KM curves, Cox PH model
ML training	Python 3.10 (sklearn, XGBoost, LightGBM)	Mortality classification
Explainability	SHAP (TreeExplainer)	Feature attribution
Validation	5-fold stratified CV + held-out test set	Performance estimation
Report	Quarto	Reproducible R + Python document

Report generated with Quarto. All code is available in the project repository. Run `quarto render report.qmd` to regenerate.